

Artwork: Test Image100

Scale-Invariant Coordinate System

$$\tilde{X} = 2 \left(\frac{x}{W} \right) - 1, \quad \tilde{Y} = 1 - 2 \left(\frac{y}{H} \right) \quad X = \frac{\left(\tilde{X} - \frac{8}{71} \right)}{\frac{146}{135}}, \quad Y = \frac{\left(\tilde{Y} - \frac{1}{86} \right)}{\frac{97}{115}}, \quad \theta = \arctan \left(\frac{Y}{X} \right)$$

Layer Definitions Boundary Manifolds

$$C_0(X, Y, v) = \frac{19 - 3v - v^2}{40} \cdot \left(1 + \frac{13}{119}X - \frac{47}{148}Y \right)$$

$$X_1 = \left(X + \frac{58}{167} \right), \quad Y_1 = \left(Y - \frac{33}{106} \right)$$

$$\tau_1(X, Y) = \frac{3}{157} \cos^2 \left(\frac{639}{40}X_1 + \frac{1153}{196}Y_1 - \frac{34}{149} \cos \left(\frac{761}{183}X_1 + \frac{217}{101}Y_1 \right) \right) + \frac{14}{195} \cos^4 \left(\frac{278}{45}X_1 + \frac{3371}{167}Y_1 + \frac{15}{88} \cos \left(\frac{372}{181}X_1 + \frac{543}{109}Y_1 \right) \right) + \frac{5}{121} \cos^6 \left(\frac{5103}{181}X_1 + \frac{1781}{146}Y_1 - \frac{14}{71} \cos \left(\frac{761}{183}X_1 + \frac{217}{101}Y_1 \right) \right)$$

$$X_2 = \left(X - \frac{8}{181} \right), \quad Y_2 = \left(Y + \frac{26}{69} \right)$$

$$\tau_2(X, Y) = \frac{39}{124} \cos^2 \left(\frac{724}{45}X_2 + \frac{857}{136}Y_2 - \frac{63}{191} \cos \left(\frac{745}{187}X_2 + \frac{266}{131}Y_2 \right) \right) + \frac{14}{59} \cos^4 \left(\frac{725}{118}X_2 + \frac{3174}{157}Y_2 - \frac{27}{155} \cos \left(\frac{280}{139}X_2 + 5.001Y_2 \right) \right) + \frac{49}{197} \cos^6 \left(\frac{4223}{150}X_2 + \frac{857}{70}Y_2 - \frac{39}{131} \cos \left(\frac{745}{187}X_2 + \frac{266}{131}Y_2 \right) \right)$$

$$X_3 = \left(X + \frac{29}{97} \right), \quad Y_3 = \left(Y - \frac{15}{109} \right)$$

$$\tau_3(X, Y) = \frac{13}{180} \cos^2 \left(\frac{1209}{74}X_3 + \frac{863}{154}Y_3 - \frac{9}{25} \cos \left(\frac{197}{54}X_3 + \frac{162}{71}Y_3 \right) \right) + \frac{5}{179} \cos^4 \left(\frac{479}{78}X_3 + \frac{1148}{57}Y_3 + \frac{9}{59} \cos \left(\frac{259}{123}X_3 + \frac{315}{59}Y_3 \right) \right) + \frac{1}{49} \cos^8 \left(\frac{1663}{137}X_3 + \frac{3169}{88}Y_3 + \frac{15}{196} \cos \left(\frac{197}{54}X_3 + \frac{162}{71}Y_3 \right) \right)$$

$$X_4 = \left(X + \frac{33}{200} \right), \quad Y_4 = \left(Y - \frac{41}{192} \right)$$

$$\tau_4(X, Y) = -\frac{1}{140} \cos^2 \left(\frac{388}{25}X_4 + \frac{11}{2}Y_4 + \frac{94}{157} \cos \left(\frac{344}{91}X_4 + \frac{395}{156}Y_4 \right) \right) - \frac{2}{171} \cos^4 \left(\frac{929}{180}X_4 + \frac{3842}{199}Y_4 \right) - \frac{2}{193} \cos^6 \left(\frac{2633}{93}X_4 + \frac{194}{17}Y_4 + \frac{49}{194} \cos \left(\frac{629}{94}X_4 + \frac{238}{95}Y_4 \right) \right) - \frac{1}{193} \cos^8 \left(\frac{2633}{93}X_4 + \frac{194}{17}Y_4 + \frac{49}{194} \cos \left(\frac{629}{94}X_4 + \frac{238}{95}Y_4 \right) \right)$$

$$X_5 = \left(X - \frac{47}{166} \right), \quad Y_5 = \left(Y + \frac{75}{193} \right)$$

$$\tau_5(X, Y) = -\frac{15}{83} \cos^2 \left(\frac{1735}{109}X_5 + \frac{866}{145}Y_5 \right) - \frac{29}{143} \cos^4 \left(\frac{659}{115}X_5 + \frac{3841}{189}Y_5 \right) - \frac{11}{75} \cos^6 \left(\frac{5495}{194}X_5 + \frac{2119}{181}Y_5 - \frac{3}{17} \cos \left(\frac{449}{77}X_5 + \frac{559}{181}Y_5 \right) \right) - \frac{11}{69} \cos^8 \left(\frac{1174}{99}X_5 + \frac{3201}{89} \cos \left(\frac{449}{77}X_5 + \frac{559}{181}Y_5 \right) \right)$$

$$X_6 = \left(X - \frac{80}{189}\right), \quad Y_6 = \left(Y + \frac{112}{141}\right)$$

$$\tau_6(X, Y) = \frac{1}{58} \cos^2 \left(\frac{2447}{164} X_6 + \frac{250}{41} Y_6 - \frac{11}{180} \cos \left(\frac{258}{59} X_6 + \frac{423}{185} Y_6 \right) \right) + \frac{5}{136} \cos^4 \left(\frac{779}{139} X_6 + \frac{2461}{123} Y_6 - \frac{81}{185} \cos \left(\frac{217}{82} X_6 + \frac{981}{193} Y_6 \right) \right) - \frac{1}{81} \cos^6 \left(\frac{1732}{61} X_6 + \frac{450}{37} Y_6 \right) - \frac{1}{108}$$

$$X_7 = \left(X + \frac{23}{99}\right), \quad Y_7 = \left(Y + \frac{79}{127}\right)$$

$$\tau_7(X, Y) = -\frac{1}{162} \cos^2 \left(\frac{3287}{195} X_7 + \frac{730}{147} Y_7 + \frac{54}{175} \cos \left(\frac{771}{166} X_7 + \frac{30}{17} Y_7 \right) \right) + \frac{5}{196} \cos^4 \left(\frac{281}{44} X_7 + \frac{2154}{107} Y_7 + \frac{9}{16} \cos \left(\frac{371}{153} X_7 + \frac{678}{101} Y_7 \right) \right) + \frac{3}{157} \cos^6 \left(\frac{3763}{135} X_7 + \frac{2107}{183} Y_7 + \frac{6}{17} \cos \left(\frac{109}{10} X_7 + \frac{109}{10} Y_7 \right) \right) - \frac{1}{108}$$

$$X_8 = \left(X + \frac{5}{134}\right), \quad Y_8 = \left(Y + \frac{5}{162}\right)$$

$$\tau_8(X, Y) = -\frac{29}{154} \cos^2 \left(\frac{2753}{173} X_8 + \frac{65}{11} Y_8 - \frac{17}{193} \cos \left(\frac{761}{190} X_8 + \frac{352}{175} Y_8 \right) \right) - \frac{39}{196} \cos^4 \left(\frac{1051}{174} X_8 + \frac{3986}{199} Y_8 \right) - \frac{33}{161} \cos^6 \left(\frac{897}{32} X_8 + 12 Y_8 - \frac{9}{107} \cos(6 X_8 + 3 Y_8) \right) - \frac{38}{199} \cos^8 \left(\frac{109}{10} X_8 + \frac{109}{10} Y_8 \right) - \frac{1}{108}$$

$$X_9 = \left(X + \frac{14}{65}\right), \quad Y_9 = \left(Y + \frac{2}{93}\right)$$

$$\tau_9(X, Y) = \frac{1}{18} \cos^4 \left(\frac{823}{144} X_9 + \frac{2918}{147} Y_9 - \frac{29}{195} \cos \left(\frac{293}{133} X_9 + \frac{161}{31} Y_9 \right) \right) - \frac{9}{107} \cos^6 \left(\frac{760}{27} X_9 + \frac{2143}{175} Y_9 \right) - \frac{23}{191} \cos^8 \left(\frac{459}{38} X_9 + \frac{5533}{153} Y_9 - \frac{16}{165} \cos \left(\frac{550}{191} X_9 + \frac{1031}{150} Y_9 \right) \right) - \frac{1}{108}$$

$$X_{10} = \left(X - \frac{16}{27}\right), \quad Y_{10} = \left(Y + \frac{19}{148}\right)$$

$$\tau_{10}(X, Y) = \frac{1}{126} \cos^2 \left(\frac{2589}{161} X_{10} + \frac{111}{20} Y_{10} - \frac{59}{191} \cos \left(\frac{282}{77} X_{10} + \frac{52}{23} Y_{10} \right) \right) - \frac{17}{169} \cos^4 \left(\frac{1084}{171} X_{10} + \frac{4028}{193} Y_{10} - \frac{116}{193} \cos \left(\frac{205}{142} X_{10} + \frac{686}{159} Y_{10} \right) \right) + \frac{1}{55} \cos^6 \left(\frac{4650}{167} X_{10} + \frac{166}{13} Y_{10} - \frac{1}{10} \cos \left(\frac{109}{10} X_{10} + \frac{109}{10} Y_{10} \right) \right) - \frac{1}{108}$$

$$X_{11} = \left(X + \frac{23}{180}\right), \quad Y_{11} = \left(Y - \frac{25}{152}\right)$$

$$\tau_{11}(X, Y) = -\frac{2}{95} \cos^2 \left(\frac{1978}{125} X_{11} + \frac{322}{53} Y_{11} + \frac{22}{183} \cos \left(\frac{813}{199} X_{11} + \frac{136}{89} Y_{11} \right) \right) + \frac{1}{76} \cos^4 \left(\frac{899}{135} X_{11} + \frac{244}{13} Y_{11} \right) - \frac{3}{64} \cos^6 \left(\frac{4408}{155} X_{11} + \frac{1343}{109} Y_{11} + \frac{95}{198} \cos \left(\frac{395}{63} X_{11} + \frac{282}{113} Y_{11} \right) \right) - \frac{1}{108}$$

$$X_{12} = \left(X - \frac{77}{178}\right), \quad Y_{12} = \left(Y + \frac{3}{61}\right)$$

$$\tau_{12}(X, Y) = -\frac{11}{167} \cos^2 \left(\frac{2698}{169} X_{12} + \frac{771}{113} Y_{12} + \frac{189}{151} \cos \left(\frac{409}{127} X_{12} + \frac{144}{121} Y_{12} \right) \right) + \frac{15}{127} \cos^4 \left(\frac{587}{120} X_{12} + \frac{2463}{130} Y_{12} - \frac{133}{200} \cos \left(\frac{547}{197} X_{12} + \frac{49}{10} Y_{12} \right) \right) - \frac{1}{184} \cos^6 \left(\frac{3361}{120} X_{12} + \frac{1}{10} \cos \left(\frac{109}{10} X_{12} + \frac{109}{10} Y_{12} \right) \right) - \frac{1}{108}$$

$$X_{13} = \left(X - \frac{13}{172}\right), \quad Y_{13} = \left(Y - \frac{31}{196}\right)$$

$$\tau_{13}(X, Y) = \frac{18}{191} \cos^2 \left(\frac{2411}{150} X_{13} + \frac{655}{129} Y_{13} + \frac{18}{167} \cos \left(\frac{18}{5} X_{13} + \frac{680}{193} Y_{13} \right) \right) - \frac{5}{186} \cos^4 \left(\frac{811}{140} X_{13} + \frac{2697}{137} Y_{13} - \frac{5}{31} \cos \left(\frac{269}{147} X_{13} + \frac{89}{18} Y_{13} \right) \right) + \frac{1}{148} \cos^6 \left(\frac{4977}{178} X_{13} + \frac{2290}{197} Y_{13} \right)$$

$$X_{14} = \left(X + \frac{11}{74}\right), \quad Y_{14} = \left(Y + \frac{21}{194}\right)$$

$$\tau_{14}(X, Y) = \frac{18}{149} \cos^2 \left(\frac{2983}{182} X_{14} + \frac{797}{195} Y_{14} + \frac{33}{145} \cos \left(\frac{749}{132} X_{14} + \frac{14}{15} Y_{14} \right) \right) + \frac{9}{182} \cos^4 \left(\frac{703}{107} X_{14} + \frac{859}{42} Y_{14} - \frac{74}{161} \cos \left(\frac{343}{198} X_{14} + \frac{208}{49} Y_{14} \right) \right) + \frac{5}{74} \cos^6 \left(\frac{4792}{169} X_{14} + \frac{2067}{178} Y_{14} \right)$$

$$X_{15} = \left(X - \frac{25}{113}\right), \quad Y_{15} = \left(Y - \frac{13}{101}\right)$$

$$\tau_{15}(X, Y) = -\frac{10}{53} \cos^2 \left(\frac{2489}{155} X_{15} + \frac{858}{161} Y_{15} - \frac{82}{125} \cos \left(\frac{262}{57} X_{15} + \frac{233}{73} Y_{15} \right) \right) + \frac{44}{197} \cos^4 \left(\frac{1038}{143} X_{15} + \frac{3121}{174} Y_{15} + \frac{53}{36} \cos \left(\frac{21}{88} X_{15} + \frac{431}{66} Y_{15} \right) \right) - \frac{5}{194} \cos^6 \left(\frac{3545}{124} X_{15} + \frac{19}{17} Y_{15} \right)$$

$$X_{16} = \left(X + \frac{83}{134}\right), \quad Y_{16} = \left(Y + \frac{67}{200}\right)$$

$$\tau_{16}(X, Y) = \frac{11}{200} \cos^8 \left(\frac{473}{20} X_{16} + \frac{3610}{167} Y_{16} \right) + \frac{6}{83} \cos^{12} \left(\frac{1097}{184} X_{16} + \frac{1643}{82} Y_{16} \right) + \frac{6}{173} \cos^{16} \left(\frac{393}{14} X_{16} + \frac{591}{49} Y_{16} - \frac{14}{193} \cos \left(\frac{1049}{176} X_{16} + \frac{373}{125} Y_{16} \right) \right) + \frac{10}{149} \cos^{16} \left(\frac{179}{15} X_{16} + \frac{179}{15} Y_{16} \right)$$

Master Composition (Recursive Layer Synthesis)

$$H_v^{(0)}(X, Y) = C_0(X, Y, v)$$

$$H_v^1(X, Y) = (1 - W_1(X, Y)) \cdot H_v^0(X, Y) + W_1(X, Y) \cdot C_1(X, Y, v)$$

$$H_v^2(X, Y) = (1 - W_2(X, Y)) \cdot H_v^1(X, Y) + W_2(X, Y) \cdot C_2(X, Y, v)$$

$$H_v^3(X, Y) = (1 - W_3(X, Y)) \cdot H_v^2(X, Y) + W_3(X, Y) \cdot C_3(X, Y, v)$$

$$H_v^4(X, Y) = (1 - W_4(X, Y)) \cdot H_v^3(X, Y) + W_4(X, Y) \cdot C_4(X, Y, v)$$

$$H_v^5(X, Y) = (1 - W_5(X, Y)) \cdot H_v^4(X, Y) + W_5(X, Y) \cdot C_5(X, Y, v)$$

$$H_v^6(X, Y) = (1 - W_6(X, Y)) \cdot H_v^5(X, Y) + W_6(X, Y) \cdot C_6(X, Y, v)$$

$$H_v^7(X, Y) = (1 - W_7(X, Y)) \cdot H_v^6(X, Y) + W_7(X, Y) \cdot C_7(X, Y, v)$$

$$\begin{aligned}
H_v^8(X, Y) &= (1 - W_8(X, Y)) \cdot H_v^7(X, Y) + W_8(X, Y) \cdot C_8(X, Y, v) \\
H_v^9(X, Y) &= (1 - W_9(X, Y)) \cdot H_v^8(X, Y) + W_9(X, Y) \cdot C_9(X, Y, v) \\
H_v^{10}(X, Y) &= (1 - W_{10}(X, Y)) \cdot H_v^9(X, Y) + W_{10}(X, Y) \cdot C_{10}(X, Y, v) \\
H_v^{11}(X, Y) &= (1 - W_{11}(X, Y)) \cdot H_v^{10}(X, Y) + W_{11}(X, Y) \cdot C_{11}(X, Y, v) \\
H_v^{12}(X, Y) &= (1 - W_{12}(X, Y)) \cdot H_v^{11}(X, Y) + W_{12}(X, Y) \cdot C_{12}(X, Y, v) \\
H_v^{13}(X, Y) &= (1 - W_{13}(X, Y)) \cdot H_v^{12}(X, Y) + W_{13}(X, Y) \cdot C_{13}(X, Y, v) \\
H_v^{14}(X, Y) &= (1 - W_{14}(X, Y)) \cdot H_v^{13}(X, Y) + W_{14}(X, Y) \cdot C_{14}(X, Y, v) \\
H_v^{15}(X, Y) &= (1 - W_{15}(X, Y)) \cdot H_v^{14}(X, Y) + W_{15}(X, Y) \cdot C_{15}(X, Y, v) \\
H_v^{16}(X, Y) &= (1 - W_{16}(X, Y)) \cdot H_v^{15}(X, Y) + W_{16}(X, Y) \cdot C_{16}(X, Y, v) \\
H_v(X, Y) &= H_v^{16}(X, Y)
\end{aligned}$$

Final RGB Channel Evaluation

$$\text{Color}(x, y) = (F(H_0(X, Y)), F(H_1(X, Y)), F(H_2(X, Y)))$$

where $F(x) = \left[255 e^{-e^{-1000x}} |x| e^{-e^{1000(x-1)}} \right]$ performs exact differentiable color clamping to $[0, 255]$.